Additional file 7　The study was divided into two groups for meta analysis according to whether EDTA was used.

| Author | Groups of Study(n) | Irrigation | included in meta analysis |
|---|---|---|---|
| Jadhav et al. (2012)[12] | BC (10)<br>BC+PRP (10) | 2.5% NaOCl | √ |
| Narang et al. (2015)[15] | BC (5)<br>PRP+Collagen (5)<br>PRF (5) | 2.5% NaOCl | √ |
| Sharma et al. (2016）[32] | BC (4)<br>PRF (4)<br>BC+Collagen (4)<br>BC+PLGA (4) | 2.5% NaOCl | √ |
| Shivashanker et al. (2017)[33] | BC(20)<br>PRP(20)<br>PRF(20) | 5.25% NaOCl | √ |
| Mittal et al. (2019) | BC(4)<br>BC+PRF (4) | 2.5% NaOCl | √ |
| Ragab et al. (2019)[35] | BC(11)<br>BC+PRF (11) | 5% NaOCl<br>saline. | Data is quantitative analysis and cannot be combined with other studies |
| Alagl et al. (2017)[16] | BC (15)<br>PRP (15) | 1.5% NaOCl<br>2.5% NaOCl<br>0.12% CHX | √ |
| Lv et al. (2018)[41] | BC (5)<br>PRF (5) | 1% NaOCl<br>Saline<br>17% EDTA | √ |
| Bezgin et al. (2015)[14] | BC (10)<br>PRP (10) | 2.5% NaOCl<br>Saline<br>0.12% CHX<br>5% EDTA | √ |
| Ulusoy et al. (2019)[39] | BC (22)<br>PRP (22)<br>PRF (22)<br>PP(22) | 1.25% NaOCl<br>2% CHX<br>17% EDTA. | √ |
| ElSheshtawy et al. (2020)[36] | PRP (14)<br>BC (17) | 5.25% NaOCl<br>17% EDTA | |
| Rizk et al. (2020)[40] | PRP (13)<br>BC+collagen (13) | 2% NaOCl<br>17% EDTA | |
| Ramachandran et al. (2020)[37] | BC+PRP+collagen (20)<br>BC (20) | 5.25% NaOCl<br>1% NaOCl<br>sterile water<br>17% EDTA | Data is quantitative analysis and cannot be combined with other studies |
| Cheng et al. (2022)[42] | BC (32)<br>CGF (30) | 0.5-1.5 % NaOCl<br>saline<br>17% EDTA | |
| Meschi et al. (2021)[43] | BC (18)<br>BC+L-PRF (11) | 1.5% NaOCl<br>Saline<br>17% EDTA | |
| Uppala et al. (2020)[38] | PRF(8)<br>BC(8)<br>BC+collagen (8) | NR | |

Four studies were included in the EDTA group[Alagl et al. (2017)[16], Lv et al. (2018)[41], Bezgin et al. (2015)[14, Ulusoy et al. (2019)[39]], and only one study was included in the PRF treatment subgroup[Ulusoy et al. (2019)[39]], so the meta-analysis could not be conducted.